2532 Ruthenium 106 Plaque Brachytherapy in Ocular Tumors

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Purpose/Objective(s): To evaluate Ruthenium 106 (Ru-106) plaque brachytherapy in the management of intraocular and adnexal tumors.

Materials/Methods: A retrospective review of 84 patients of Ocular Tumors treated with Ru-106 Plaque Brachytherapy at the L.V.P.E. Institute (Jan' 2001 - Aug' 2009).

Results: The tumors included uveal melanoma (28), ocular surface squamous neoplasia (19), choroidal hemangioma (19), retinoblastoma (15) and choroidal metastasis (2). Uveal melanoma: A notch plaque was used in 71%. The mean tumor apex dose was 9528 cGy. Tumor regression rate was 67.8%, eye salvage was achieved in 82%, and 71.42% had useful residual vision (≥20/200). Ocular surface squamous neoplasia (n = 19): The mean tumor apex dose was 5626 cGy. Tumor regression rate was 84.21%, eye salvage was achieved in 78%, and 63.15% had useful residual vision (≥20/200). Choroidal hemangioma (n = 19): The mean tumor apex dose was 3246 cGy. Tumor regression rate was 89.6%, eye salvage was achieved in all and 57.9% had improvement in visual acuity (>2 Snellen lines). Retinoblastoma (n = 15): The mean tumor apex dose was 4699 cGy. Tumor regression rate was 68.75%, eye salvage was achieved in 60%, and 33.3% had useful residual vision (≥20/200). Choroidal metastasis (n = 2): Plaque brachytherapy was used to treat solitary choroidal metastasis. The mean tumor apex dose was 4995 (range, 4992 - 4998) cGy. Tumor regression was seen in both, eye salvage was achieved in 100%, and 100% had useful residual vision (≥20/200).

Conclusions: Ruthenium 106 plaque brachytherapy is a reasonable treatment option for the primary management of choroidal melanoma and choroidal hemangioma with diffuse subretinal fluid, and for the management of residual or recurrent ocular surface squamous neoplasia and retinoblastoma. It provides for good tumor regression and eye salvage. Complications seem dose-dependent.

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2533 Predicting Outcomes after Salvage Reirradiation of Recurrent Head and Neck Cancer

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Purpose/Objective(s): The management of recurrent head and neck cancer (HNC) where patients received prior HN radiation is a challenging clinical situation. Trotti *et al* established a nomogram predicting for 24-month overall survival (OS) in this patient group, based on comorbidity, organ dysfunction, isolated neck recurrence, tumor bulk and time interval since prior RT. We sought to validate this nomogram and also develop a nomogram for locoregional failure (LRF), an important endpoint in recurrent HNC.

Materials/Methods: The records were reviewed for all 105 patients with locally recurrent HNC without evidence of distant metastasis who underwent reirradiation at MSKCC between 7/96 and 9/05. Comorbidity was assessed by Charlson index. Organ dysfunction was defined as feeding tube dependency, functioning tracheostomy or soft tissue defect. OS was estimated by Kaplan-Meier methods. Factors were included in the multivariate model only if significant on univariate analysis. Each nomogram was constructed from the multivariate model and concordance probability was produced to assess the model performance.

Results: Patient/disease characteristics: 80% age <70; 66% male; 81% non-nasopharynx; 87% SCC. The median interval since prior radiation was 4.6 years. Recurrent disease prior to reirradiation was staged as rT4 in 66% of patients and N+ in 45% of patients. Regarding their reirradiation course, 76% of patients received >50 Gy and 70% were treated with IMRT. Prior to reirradiation, 32% underwent surgical resection and 71% received concurrent chemotherapy with reirradiation. By Charlson index, 36 pts had an index ≥1 excluding cancer. Organ dysfunction was noted in 56 pts. With a median follow-up of 48 months for surviving patients, the 5 year OS rate was 25%. Two year cumulative incidence of LRF was 0.46. A number of factors were highly significant for OS on univariate, but not multivariate, analysis. These variables, including Charlson index, organ dysfunction and isolated neck recurrence, were previously incorporated into the Trotti OS nomogram. Four different factors predicted OS - Karnofsky performance scale (KPS), Nasopharynx (NPX) primary, SCC histology and surgery prior to reirradiation. The concordance index (CI) was 0.75. Three factors independently predicted incidence of LRF: tumor stage, IMRT, and organ dysfunction (CI 0.66).

Conclusions: Our study highlights four criteria for predicting 2 year OS: 1) KPS<80, 2) NPX, 3) SCC and 4) surgery. Factors included in our OS nomogram are distinct from that previously published. Our competing risks nomogram predicting cumulative incidence of LRF at 2 years is novel, with rT4, IMRT, and organ dysfunction as significant predictors of outcome. Future work will evaluate patients treated since 9/05 for further refinement and validation of these nomograms.

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2534 Characteristics and Clinical Outcomes of Patients with Multiple Potentially HPV-Related Malignancies

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Purpose/Objective(s): Human papillomavirus (HPV) is associated with the development of cervical, vulvar, vaginal, anal, and penile cancers, as well as a large subset of head and neck squamous cell cancers (HNSCC). It is unknown whether the development of one potentially HPV-related cancer affects the development of another and how the disease characteristics and outcomes are related. To address these issues, we identified a cohort of patients with multiple potentially HPV-related malignancies and examined the tumor characteristics and patient outcomes.